

SignSight – Intelligent Tamil Sign Language System

Project Overview

SignSight is an AI-powered, end-to-end **Tamil Sign Language (TSL) learning, translation, and emotional support platform** designed to bridge communication gaps for the deaf and hearing-impaired community in Sri Lanka.

The system integrates **computer vision, deep learning, explainable AI, emotion analysis, and personalized learning** to support children, learners, guardians, and mentors. It moves beyond basic sign recognition by offering **emotion-aware child monitoring, dynamic sign detection, AI-driven evaluation, mentorship, and audio-to-sign translation** in a single unified ecosystem.

System Architecture Overview

SignSight consists of **four tightly integrated components**, each developed by a team member, connected via a centralized backend and shared data flow.

Core Components

1. Emotion & Context-Aware Child Module
 2. Personalized Learning, Evaluation & Mentorship Module
 3. Dynamic Tamil Sign Detection Module
 4. Audio-to-Sign Language Video Conversion Module
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Component 1: Emotion & Context-Aware Child Module

Purpose

Designed for **children aged 3–10**, this module analyzes **facial emotional reactions** while children watch cartoon content. It helps guardians understand emotional well-being and learning comfort in real time.

Key Features

- Real-time emotion detection: **Happy, Sad, Angry, Fear, Neutral**
- Webcam-based facial analysis
- Guardian email reports with emotional insights and parenting tips
- Visual emotion summaries and confidence scores

Technologies

- MobileNetV2 (TensorFlow/Keras)
- MediaPipe Face Detection
- OpenCV
- Flask (Backend)
- React + Vite + Tailwind CSS
- SMTP / Flask-Mail

Emotion Flow

1. Child watches curated cartoon videos
2. Webcam captures facial reactions
3. AI model predicts emotions in real time
4. Emotion summary + recommendations sent to guardian

Novelty

- First system combining **emotion detection with Tamil Sign Language learning**
- Enables **non-verbal emotional monitoring** for children
- Enhances parental awareness and child-centered education

GitHub Repository

- <https://github.com/GK-NETHMI/SignSight-Frontend.git>
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Component 2: Personalized Learning, Evaluation & Mentorship Module

Purpose

Provides **adaptive learning paths, intelligent evaluation, explainable AI feedback, and mentor-guided supervision** to ensure fair and effective learning.

Core Features

- Personalized lesson recommendations
- ML-based performance prediction (0–100)
- Explainable AI using SHAP
- Suspicious behavior detection in quiz videos
- Mentor monitoring and feedback system

Technologies

- Flask (Python)

- Scikit-learn (Random Forest Regressor)
- SHAP (Explainable AI)
- NumPy, Pandas
- OpenCV
- React + Vite + Tailwind CSS

Workflow

1. Learner completes lessons and quizzes
2. Performance data collected
3. ML predicts performance score
4. SHAP explains strengths and weaknesses
5. Personalized recommendations generated
6. Mentor reviews flagged attempts

Novelty

- Combines **AI + human mentorship**
- Transparent evaluation through Explainable AI
- Ensures fairness and academic integrity

GitHub Repository

- <https://github.com/GK-NETHMI/SignSight-Frontend.git>
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Component 3: Dynamic Tamil Sign Detection System

Purpose

Recognizes **dynamic and continuous Tamil Sign Language gestures** using real-time webcam input or offline video uploads.

Key Features

- Real-time webcam sign detection
- Offline MP4 video-based sign recognition
- Dynamic gesture recognition using motion sequences
- Confidence scoring and landmark visualization

Technologies

- LSTM (Deep Learning Model)
- MediaPipe Hands
- OpenCV
- Flask (Backend API)
- React + Vite + Tailwind CSS

Workflow

1. User selects webcam or video upload
2. Hand landmarks extracted
3. Motion sequences encoded
4. LSTM predicts dynamic sign
5. Result displayed with confidence score

Novelty

- Focuses on **dynamic gestures**, not static poses
- Supports real-world offline video analysis
- Tailored specifically for Tamil Sign Language

GitHub Repository

- <https://github.com/ff786/Sign-Sight-Application-FE.git>
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Component 4: Audio-to-Sign Language Video Conversion Module

Purpose

Transforms **spoken Tamil audio or video content into Tamil Sign Language videos**, enabling inclusive communication for deaf users.

Core Features

- Drag-and-drop audio/video upload (MP3, WAV, MP4)
- Speech-to-text conversion
- AI-driven sign language mapping
- Interactive video playback with controls
- Real-time conversion status tracking

Technologies

- React + Vite
- Tailwind CSS
- HTML5 Video API
- Fetch API
- Backend AI services (Speech-to-Text & Sign Mapping)

Workflow

1. User uploads audio/video file
2. Speech converted to Tamil text

3. Text mapped to sign gestures
4. Sign language video generated
5. Video delivered with playback controls

Novelty

- Converts **spoken Tamil directly into sign language**
- Improves accessibility for education and daily communication
- User-friendly and mobile-responsive design

GitHub Repository

- <https://github.com/ff786/Sign-Sight-Application-FE.git>
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Common Tech Stack & Tools

Backend & AI

- Python, Flask
- TensorFlow / Keras
- Scikit-learn
- SHAP
- OpenCV
- MediaPipe

Frontend

- React
- Vite
- Tailwind CSS
- TypeScript (TSX)

Development Tools

- Git & GitHub
 - VS Code
 - Postman
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How to Run (General)

Backend

```
cd backend  
pip install -r requirements.txt  
python app.py
```

Frontend

```
cd frontend
npm install
npm run dev
```

API Overview (Sample)

Endpoint	Purpose
/api/evaluate-quiz	Learner evaluation & XAI feedback
/api/convert-to-sign	Audio/Video to sign conversion
/api/detect-emotion	Child emotion analysis
/api/detect-sign	Dynamic sign recognition

Overall Novelty & Impact

- First **fully integrated Tamil Sign Language AI platform**
 - Combines **emotion awareness, dynamic gesture recognition, explainable learning, mentorship, and translation**
 - Designed for **real-world deployment in Sri Lanka**
 - Empowers children, learners, parents, mentors, and educators
 - Promotes inclusivity, transparency, and accessibility in education
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Conclusion

SignSight transforms Tamil Sign Language learning and communication from a fragmented experience into a **smart, emotional, explainable, and inclusive AI-driven ecosystem**, setting a strong foundation for future assistive technologies in Sri Lanka and beyond.

TAMIL SIGN LANGUAGE PLATFORM – SYSTEM ARCHITECTURE

An AI-powered platform for translation, understanding, learning and intelligent analysis of Tamil Sign Language.

